Alberto Salvarese

alberto.salvarese@utexas.edu Alberto Salvarese

Education

Aug. 2022 – Present
Ph.D. University of Texas at Austin, Physics.
Research interests: Gravitational wave cosmology and astrophysics; Bayesian statistical inference; Population studies
Supervisor: Professor Hsin-Yu Chen
GPA: 3.98

Oct. 2019 – Apr. 2022
M.Sc. Università degli studi di Padova, Astrophysics and Cosmology
Thesis title: *Probing parity violation in the Early Universe*.
Supervisor: Professor Nicola Bartolo (Co-supervisor: Doctor Alessandro Greco)
Final grade: 110/110 cum laude

Oct. 2016 – Sept. 2019
B.Sc. Università degli studi di Padova, Astronomy
Thesis title: *Constraints on the Neutron Stars equation of state from multiwavelength observations of the event GW170817*.
Supervisor: Professor Roberto Turolla
Final grade: 107/110

Research

First Author Journal articles

- A. Salvarese, H.-Y. Chen, A. Zimmerman, *et al.*, "Eccentricity as a Magnifying Glass: Precision Population Inference Enabled by Eccentric Neutron Star–Black Hole Mergers," Jul. 2026. arXiv: 2607.16136 [astro-ph.HE]. URL: https://arxiv.org/pdf/2607.16136
- A. Salvarese, H.-Y. Chen, and D. E. Holz, *Identifying the host of compact binary mergers*, 2026. arXiv: 2604.28132 [astro-ph.CO]. URL: https://arxiv.org/abs/2604.28132
- A. Salvarese, H.-Y. Chen, A. Mangiagli, *et al.*, "Listening across the cosmic time: Standard sirens from ground- and space-based missions in the next decade," *Classical and Quantum Gravity*, vol. 42, no. 19, p. 195 002, Sep. 2025. DOI: 10.1088/1361-6382/ae05ab
- A. Salvarese and H.-Y. Chen, "Mitigating the binary viewing angle bias for standard sirens," *The Astrophysical Journal Letters*, vol. 974, no. 1, p. L16, Oct. 2024. DOI: 10.3847/2041-8213/ad7bbc

Other Journal articles

- LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Gwtc-5.0: Constraints on the cosmic expansion rate and modified gravitational-wave propagation," 2026. arXiv: 2605.27227 [astro-ph.CO]. URL: https://arxiv.org/abs/2605.27227
- LIGO Scientific Collaboration, Virgo Collaboration, and KAGRA Collaboration, "Gwtc-4.0: Constraints on the cosmic expansion rate and modified gravitational-wave propagation," 2025. arXiv: 2509.04348 [astro-ph.CO]. URL: https://arxiv.org/abs/2509.04348

Conferences and talks

- *American Physical Society Global Summit*, March 2026, Denver, Colorado; in-person speaker
- *American Physical Society Global Summit*, March 2025, Anaheim, California; in-person speaker
- *EDSU Tools 2024 - 5th World Summit*, June 2024, Île de Noirmoutier, France; accepted speaker

- *American Physical Society April meeting*, April 2024, Sacramento, California; in-person speaker

Other projects

- *Measuring the thermal conductivity of 6061-T6 aluminum*, University of Texas at Austin, Supervisor: Professor N. Galitzki, Summer 2024 (H. E. Nachman *et al.*, Sep. 2025. arXiv: 2509.23422 [astro-ph.IM])
- *Analysis of the AGN NGC2110*, Università degli studi di Padova, Supervisor: Professor E. Bottacini, Fall 2019

Awards and Fellowships

2026–2027 Distinguished Graduate Fellowship in Physics, The University of Texas at Austin

Workshops

Jun. 2026 "AI Boot Camp for Astronomers: Hands-on AI training for the next generation of astronomy research", Austin, Texas, June 1-5 2026

Jan. 2026 "GWfreeride: Carving the AI Gradient in Gravitational-Wave Astronomy", Sexten, Italy, January 26-30 2026

Jul. 2025 "Prospects in Theoretical Physics 2025", Institute for Advanced Studies (IAS), July 14-25 2025

Jun. 2025 "Caltech Relativistic Astrophysics Summer School 2025", Caltech, June 16 - 20 2025.

Aug. 2021 "FERMILAB 2021 summer student school at LNF", Laboratori Nazionali di Frascati.

Nov. 2020 "Statistical Methods for the Detection, Classification, and Inference of Relativistic Objects", ICERM Brown University, November 16 - 20 2020.

Skills

Coding Python; R; LaTeX; basics of Mathematica; basics of SQL

3D modelling basics of SOLIDWORKS